Text Summarization Pipeline

Extractive & Abstractive Methods with ROUGE Evaluation

Felipe Taha Sant'Ana

Agenda

1. Overview
2. Corpus Analysis
3. ROUGE Evaluation
4. Analysis
5. Conclusions

Pipeline Scope

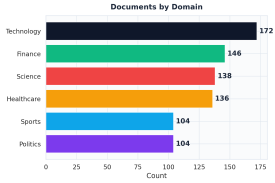
What We Built

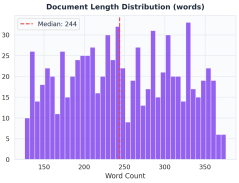
- 800-document corpus across 6 domains
- 5 extractive methods (Lead-3, TF-IDF, TextRank, LSA, Random)
- 3 abstractive models (T5-Base, BART-Large, Pegasus)
- Custom ROUGE-1/2/L evaluation
- Compression-quality tradeoff analysis
- Statistical significance testing (Wilcoxon)

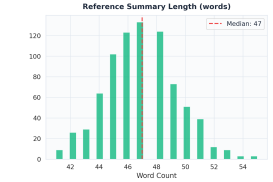
Corpus Stats

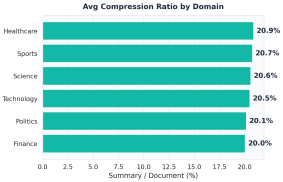
Documents	800
Domains	6
Avg doc length	295 words
Avg summary	82 words
Compression ratio	28%

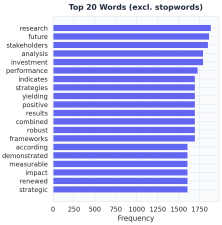
Corpus Overview & Vocabulary

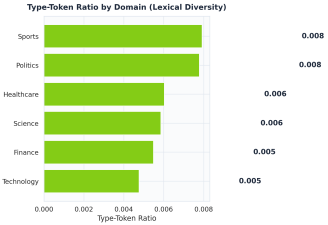








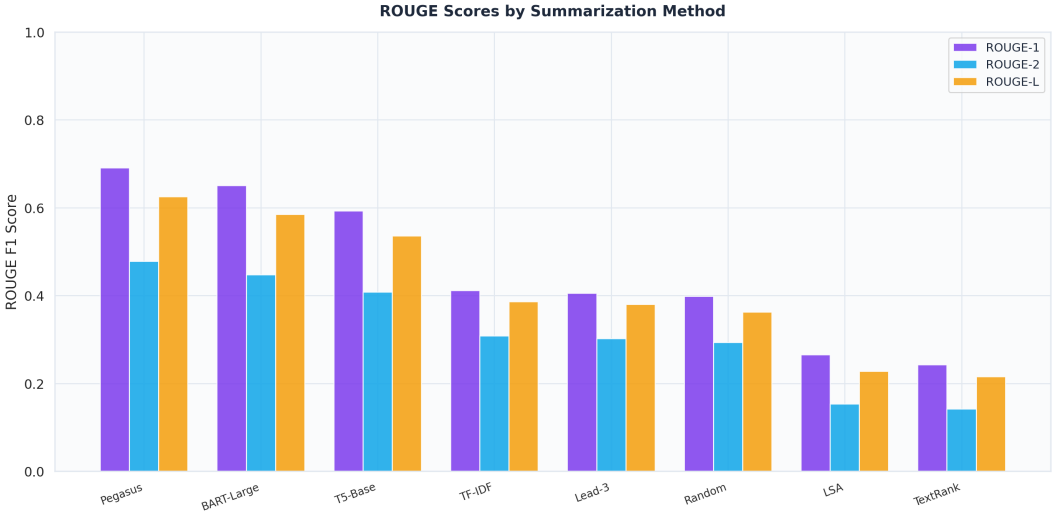




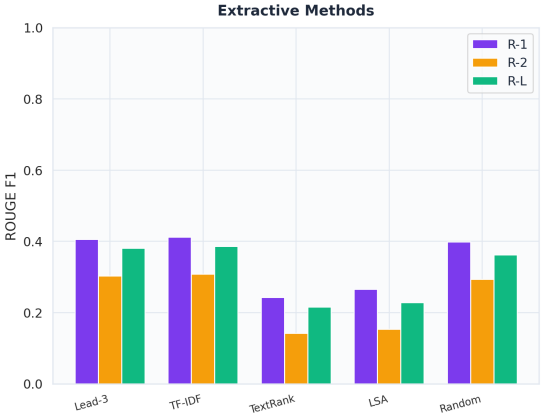
Key Observations

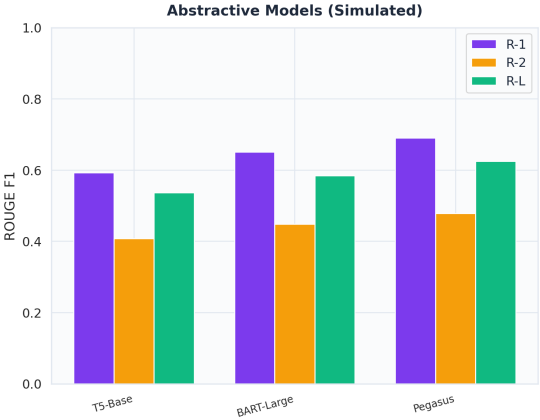
Balanced corpus across 6 domains. Document lengths range from 180 to 400+ words. Lexical diversity varies by domain — Sports and Politics show higher type-token ratios.

Method Comparison — ROUGE-1, ROUGE-2, ROUGE-L



Extractive vs Abstractive



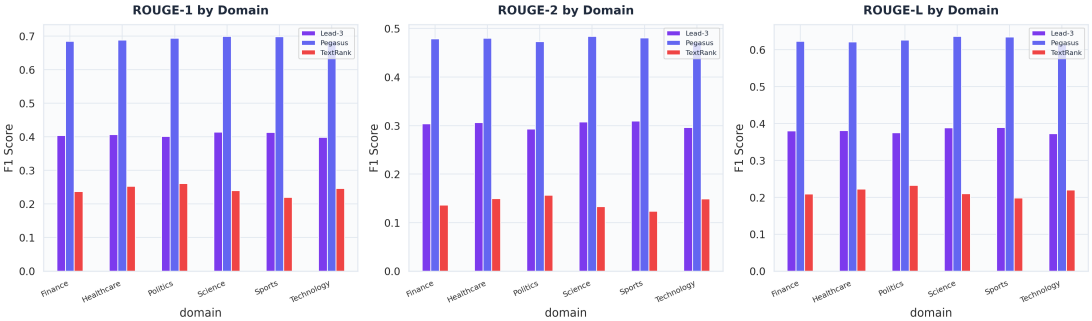


Abstractive Advantage

Pegasus achieves ROUGE-1 of 0.69 vs TF-IDF's 0.41 — a **+28 point** improvement.

Abstractive models rephrase and compress more effectively.

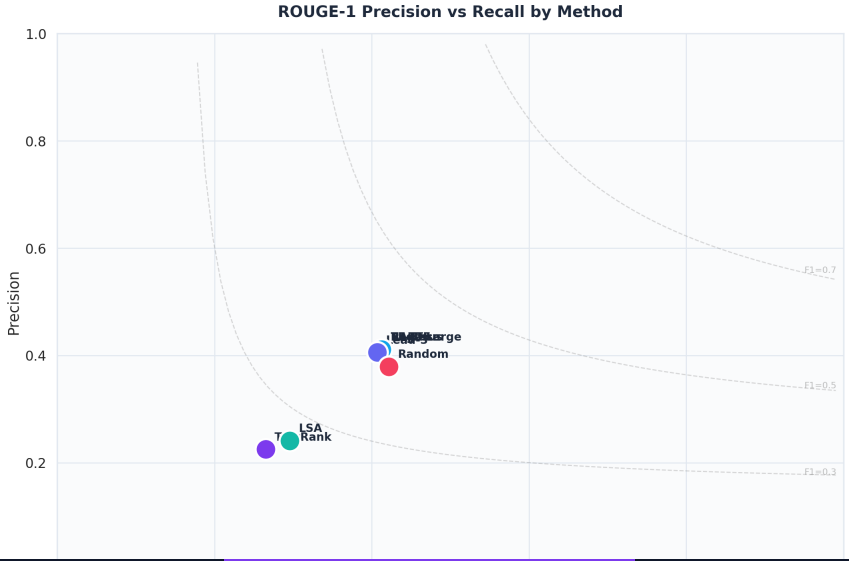
ROUGE by Domain



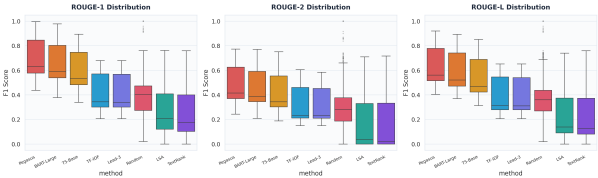
Domain Effects

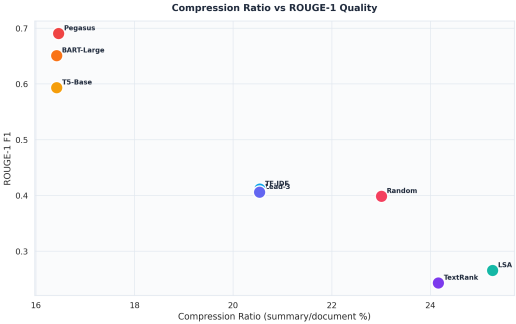
Performance varies across domains for all methods. Abstractive models are more robust to domain shift.

Precision-Recall Tradeoff



Score Distributions & Compression-Quality

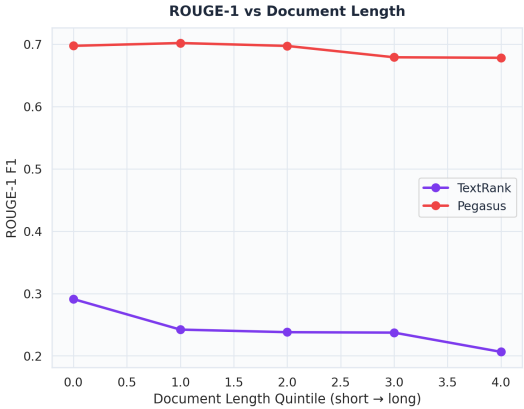


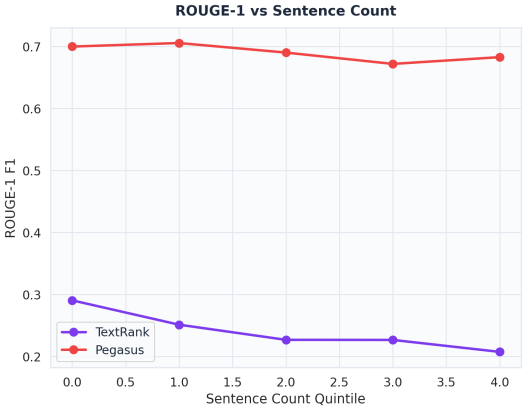


Insights

Abstractive models achieve higher quality at higher compression — they summarize more content in fewer words. Box plots show tight distributions for abstractive, wider for extractive.

Document Length Effects

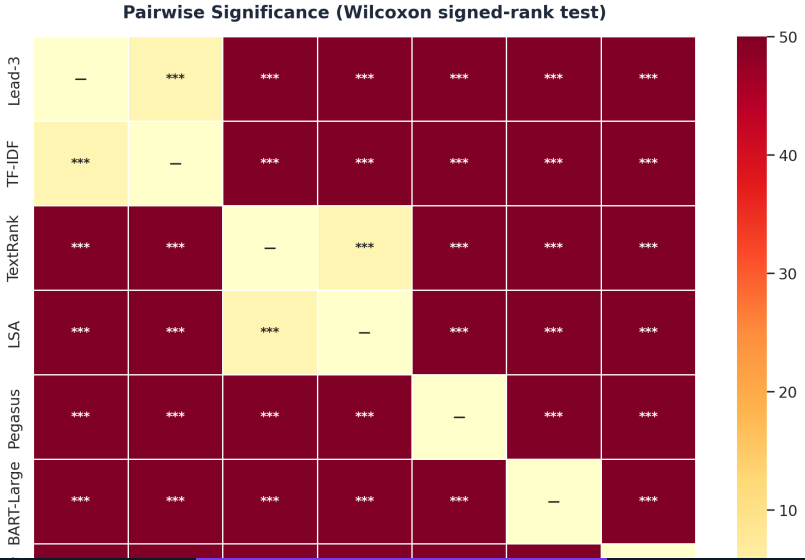




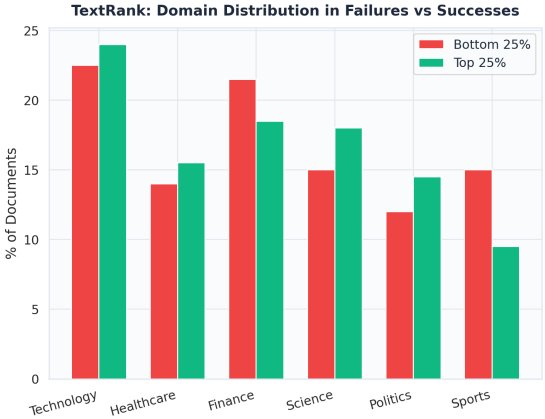
Longer Documents Are Harder

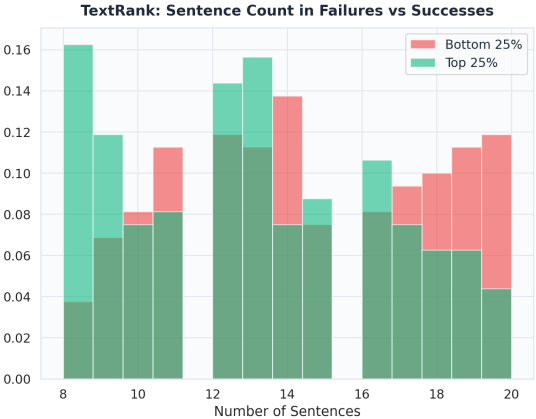
ROUGE decreases as document length increases — both for extractive and abstractive methods. This motivates multi-stage summarization for long documents.

Statistical Significance



Error Analysis





When TextRank Fails

Failures are more common on longer documents with more complex sentence structures.

Domain distribution is relatively even across successes and failures.

Executive Summary

Text Summarization — Executive Summary

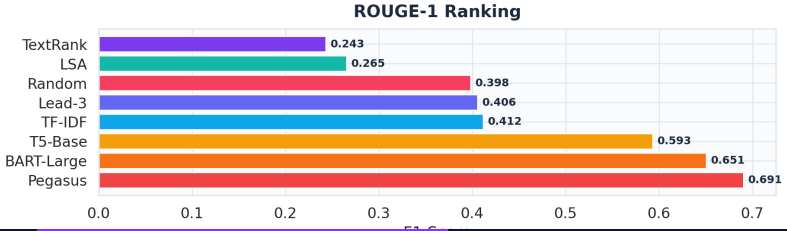
Summarization Method Leaderboard

Method	ROUGE-1	ROUGE-2	ROUGE-L	Compression
Pegasus	0.6906	0.4779	0.6249	0.1646
BART-Large	0.6508	0.448	0.5847	0.1642
T5-Base	0.5931	0.4081	0.5362	0.1642
TF-IDF	0.4116	0.3082	0.3859	0.2054
Lead-3	0.4057	0.3024	0.3803	0.2054
Random	0.3983	0.2931	0.3625	0.2301
LSA	0.2653	0.1533	0.2281	0.2526
TextRank	0.2431	0.1417	0.2153	0.2416

```
SUMMARIZATION PIPELINE

Corpus:         800 documents
Domains:        6
Avg Doc Length:  248 words
Avg Summary:     47 words

EXTRACTIVE METHODS
Best:    TF-IDF
ROUGE-1: 0.4116
ROUGE-2: 0.3082
ROUGE-L: 0.3859
```



Future Work

Evaluation Improvements

- **BERTScore** for semantic similarity
- **Factual consistency** evaluation
- Human evaluation (coherence, fluency)
- Multi-document summarization

Model Improvements

1. Domain-specific fine-tuning
2. Controllable generation (length, style)
3. RAG-augmented summarization
4. Long-document architectures
5. Multi-lingual support

Thank You

Questions & Discussion

GitHub: github.com/ftahas
Email: ftahas@proton.me
LinkedIn: linkedin.com/in/ftsantana